	DEPARTMENT OF MATHEMATICS **Academic Year:2025-2026**	**Submission Date:** **02/02/2026** **Max. Marks:50**
	B.E/B.Tech : **Specialisation:** **Student Department:CSE/INT** **Semester: IV** **Assignment No. : I** **Sub. Code & Name : MA22454-Probability and Queueing Theory**	

Q.No	Question / Task / Activity	CO*	RBT Level#	Marks
1	Is the function defined as follows p.d.f.? $f(x) = \begin{cases} 0, if \quad x < 2 \\ \frac{3+2x}{18} \; if \quad 2 \le x \le 4 \\ 0, if \quad x > 4 \end{cases}$. If so find $P(2 \le x \le 3)$.	**1**	**AP**	**10**
2	The amount of time in hours that a computer functions before breaking down is a continuous random variable with probability density function given by $f(x) = \begin{cases} \lambda e^{-\frac{x}{100}}, x \ge 0 \\ 0, \quad x < 0 \end{cases}$.What is the probability that (a) A computer will function between 50 and 150 hours before breaking down (b) It will function less than 500 hrs.	**1**	**AP**	**20**
3	Find the mean and variance of the continuous random variable X if it has the density function $f(x) = \begin{cases} 2(x-1), \quad 1 < x < 2 \\ 0, \quad otherwise \end{cases}$.	**1**	**AP**	**10**
4	If the p.d.f. of X is given by $f(x) = \begin{cases} 2(1-x), \quad 0 < x < 1 \\ 0, \quad otherwise \end{cases}$. (a) Show that $E(X^r) = \frac{2}{(r+1)(r+2)}$. (b) Using this result, evaluate $E[(2X+1)^2]$.	**1**	**AP**	10
Total Marks				**50**